TRIBHUVAN UNIVERSITY
INSTITUTE OF SCIENCE AND TECHNOLOGY
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN

A Internship Report On

Design and Implementation of a Secure Full Stack Content Management System (CMS) for Chuirika Using the PERN Stack

Submitted To:
DEPARTMENT OF IT
CENTRAL CAMPUS OF TECHNOLOGY (CCT), HATTISAR, DHARAN, SUNSARI, KOSHI, NEPAL

In partial fulfilment of the requirements for the degree of Bachelor's of Science in Computer Science and Information Technology (B.Sc. CSIT)

Submitted By:
DARSHAN DHAKAL [29592/078]
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Under the Supervision of:
Mr. Prakash Neupane

Falgun, 2082

INTERNSHIP CERTIFICATION

Regd. No. 149512/72/073
Information Care Pvt. Ltd.
Ward No. 10, Mahadevsthan Marg, Kathmandu

February 5, 2026

To Whom It May Concern

Subject: Internship Completion Confirmation

This letter is to inform you that **Mr. Darshan Dhakal** has successfully completed his Software Development Internship at **Information Care Pvt. Ltd.** as a **Full Stack Web Developer** from **November 2, 2025** to **February 2, 2026**.

During the internship period, he worked with and on both frontend and backend web technologies like React.js, Node.js, Express.js, PostgreSQL, Restful APIs, GraphQL. He was involved in developing user interfaces, implementing server-side functionality, developing and integrating APIs, and working with database models, relationships and queries. He performed the assigned duties sincerely and responsibly.

Throughout the internship, he showed good technical understanding, a positive learning attitude, and professionalism in the workplace. His conduct and performance were highly satisfactory during the entire internship period.

We appreciate his contribution and wish him all the best for future academic and professional endeavours.

Sincerely,

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SUPERVISOR RECOMMENDATION

This is to recommend that **DARSHAN DHAKAL [29592/078]** has carried out Internship on the position of "**Software Development**" in partial fulfilment of the requirements for the degree of Bachelor of Science (B.Sc.) degree in CSIT under my supervision in the Department of Information Technology, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

To my knowledge, this work has not been submitted for any other degree. The students have fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the submission of the project work for the partial fulfillment of Bachelor of Science (B.Sc.) degree in CSIT

Mr. Prakash Neupane
Supervisor
Department of IT
Central Campus of Technology
Hattisar, Dharan, Sunsari, Koshi, Nepal

CERTIFICATE OF APPROVAL

This is to certify that the internship report entitled "**Software Development** ", submitted by **Mr. DARSHAN DHAKAL [29592/078]**, a student enrolled in the Bachelor of Science (B.Sc.) degree in CSIT at the Central Campus of Technology, Dharan, has been examined and approved by the undersigned.

The report has been evaluated based on its relevance to the internship experience, presentation of facts, adherence to prescribed guidelines, and overall quality. It has been found to meet the standards set by Tribhuvan University for the partial fulfillment of the requirements for the bachelor's degree.

Asst. Prof. Narayan Dhamala	**Mr. Balabhadra Bhandari**
External Examiner	**Program Director**
Patan Multiple Campus, TU	Department of IT
Kathmandu Nepal	Central Campus of Technology, TU
	Dharan, Nepal

DECLARATION

This internship report, entitled "**Software Development**", is being submitted to the Department of IT, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal, for the partial fulfillment of the requirements for the Bachelor of Science (B.Sc.) degree in CSIT. This project has been carried out by us under the supervision of Prakash Neupane, T.U., Department of IT, Central Campus of Technology, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

We hereby declare that this work is original and has not been submitted earlier, in part or in full, to this or any other university or institution, here or elsewhere, for the award of any degree.

DARSHAN DHAKAL [29592/078]
Department of IT
Central Campus of Technology
Hattisar, Dharan, Sunsari, Koshi, Nepal

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to **Information Care Pvt. Ltd.** for providing me with the opportunity to undertake my internship as a **Full Stack Web Developer**. This internship provided valuable industry exposure and practical experience, allowing me to strengthen my technical knowledge and professional skills in a real-world working environment.

I am deeply grateful to my mentor, **Mr. Aashish Pokhrel**, for his continuous guidance, technical support, and encouragement throughout the internship period. His mentorship played a vital role in enhancing my understanding of web development and software development practices.

I would also like to extend my sincere thanks to my supervisor, **Mr. Prakash Neupane**, for his support, constructive feedback, and supervision during the internship. His guidance greatly contributed to the successful completion of my assigned tasks and projects.

Furthermore, I would like to thank the faculty members of the **Department of Computer Science and Information Technology**, **Institute of Science and Technology (IoST)**, **Tribhuvan University**, for their academic guidance and support throughout my studies.

I am thankful to my colleagues, friends, and fellow students for their encouragement and support during this internship. I am also sincerely grateful to my family for their unwavering support and motivation throughout this journey.

Finally, I would like to thank everyone who directly or indirectly contributed to making this internship a valuable and enriching learning experience.

Thank you.

DARSHAN DHAKAL [29592/078]

ABSTRACT

This report presents the work carried out during the internship as a **Full Stack Web Developer** at **Information Care Pvt. Ltd.**. The internship was conducted with the objective of gaining practical exposure to full-stack application development, system architecture design, and modern web technologies in a professional environment.

During the internship period, various frontend and backend technologies were studied and implemented to design, develop, and maintain scalable web applications. On the frontend, responsive and user-centric interfaces were developed using modern JavaScript frameworks and component-based architecture principles. On the backend, RESTful APIs were designed and implemented with proper routing, authentication, and database integration to ensure secure and efficient data management.

Major activities included designing database schemas, implementing CRUD operations, integrating third-party APIs, managing authentication and authorization mechanisms, and optimizing application performance. The internship also involved version control using Git, collaborative development workflows, debugging, and deployment of applications to cloud environments.

In addition, tasks related to testing, performance optimization, and security enhancement were performed to ensure application reliability and robustness. Emphasis was placed on writing clean, maintainable code following industry best practices and software development lifecycle methodologies.

Overall, the internship provided practical knowledge of full-stack development, system integration, and real-world software engineering practices, helping to bridge the gap between academic learning and industry-level application development.

Keywords: Full Stack Development, Web Applications, Frontend, Backend, REST API, Database, Authentication, Deployment.

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LIST OF ABBREVIATIONS

ACID Atomicity, Consistency, Isolation, Durability.

API Application Programming Interface.

CI/CD Continuous Integration and Continuous Delivery.

CMS Content Management System.

CRUD Create, Read, Update, Delete.

DOM Document Object Model.

HTTP Hypertext Transfer Protocol.

I/O Input/Output.

JWT JSON Web Token.

PERN PostgreSQL, Express.js, React.js, Node.js.

RBAC Role-Based Access Control.

REST Representational State Transfer.

SDLC Software Development Lifecycle.

SEO Search Engine Optimization.

SQL Structured Query Language.

TDD Test-Driven Development.

UI User Interface.

UX User Experience.

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CHAPTER 1
INTRODUCTION

1.1. Introduction

Throughout the internship, I was involved in developing and maintaining full stack web applications based on real client requirements. On the frontend, I built reusable React components and designed responsive user interfaces to improve user experience. On the backend, I developed RESTful APIs using Node.js and Express.js (Fielding, 2000), implemented authentication mechanisms using JWT (Jones, 2015), and handled database interactions through structured queries in PostgreSQL.

The rapid growth of web technologies has significantly transformed how modern software applications are designed and developed. Today's applications are expected to be interactive, scalable, secure, and responsive across different devices. This shift has increased the demand for full stack developers who can work on both the frontend and backend of web systems and understand how all components integrate into a complete application.

Full stack development allows a developer to handle the entire workflow of a web application, from designing user interfaces to building server-side logic and managing databases. During my internship at **Information Care Pvt. Ltd.**, I worked as a **Software Development Intern (Full Stack Web Developer)** where I primarily used the PERN stack (PostgreSQL, Express.js, React.js, and Node.js). This stack enabled me to work with JavaScript across the application while also gaining experience in relational database management using PostgreSQL.

One of the most valuable aspects of this internship was understanding how different layers of an application communicate with each other. I learned how to structure backend projects properly, organize routes and controllers, manage middleware, design normalized database schemas (Codd, 1970), and ensure secure data transmission between client and server (McGraw, 2006). Working in a collaborative environment also strengthened my skills in Git-based version control, debugging, and writing clean, maintainable code.

This internship helped me connect theoretical concepts learned in academic courses with real-world software development practices. It improved my understanding of software development lifecycle (SDLC) (Sommerville, 2010), system design principles (Newman, 2015), REST architecture (Fielding, 2000), and performance optimization in full stack applications (Beck, 2001).

1.2. Problem Statement

While developing modern web applications, several practical challenges can arise. Many traditional systems suffer from tightly coupled architectures, poor database design, and weak security mechanisms (McGraw, 2006). Without a structured development approach, applications may become difficult to maintain, scale, or extend (Sommerville, 2010).

During full stack development, common technical challenges include:

- Designing a clear separation between frontend and backend logic (Newman, 2015)
- Managing database relationships and optimizing queries (Codd, 1970)
- Implementing secure authentication and authorization mechanisms (McGraw, 2006)
- Handling asynchronous operations in backend services
- Managing application state effectively in frontend applications
- Ensuring proper validation and error handling

In PERN stack applications specifically, developers must ensure that React components efficiently consume APIs, Express routes are well-structured, PostgreSQL schemas are normalized, and Node.js handles concurrent requests properly. Without proper design practices, issues such as slow performance, redundant code, and security vulnerabilities can occur (Humble & Farley, 2010).

Therefore, it was necessary to follow a structured and practical approach to build scalable and secure full stack applications that maintain code quality and performance standards.

1.3. Objectives

The main objective of this internship was to gain practical experience in full stack web development using the PERN stack in a professional environment. The specific objectives were:

- To design and develop dynamic web applications using PostgreSQL, Express.js, React.js, and Node.js.
- To implement RESTful APIs following proper architectural practices.
- To design efficient and normalized relational database schemas.
- To implement authentication and authorization using JWT.
- To improve problem-solving, debugging, and code optimization skills.
- To understand teamwork, version control, and real-world development workflows.

1.4. Scope and Limitation

Scope

The internship mainly focused on practical full stack development tasks, including:

- Frontend development using React.js with reusable component design
- Backend API development using Node.js and Express.js
- PostgreSQL database design and query implementation
- Implementation of CRUD operations
- Authentication and authorization mechanisms
- Git-based version control and collaborative development
- Basic deployment and application testing

The experience provided exposure to real development environments where applications are built based on actual business requirements. It also improved my understanding of how to write production-level code and maintain coding standards within a team.

The internship is directly related to academic subjects such as Software Engineering, Database Management Systems, Web Technology, and Object-Oriented Programming.

Limitation

Despite the valuable learning experience, there were certain limitations:

- The internship duration was limited.
- Access to highly sensitive production data was restricted.
- Exposure to large-scale enterprise architectures was limited.
- The primary focus remained on the PERN stack rather than multiple technology stacks.

Some advanced topics such as large-scale distributed systems, enterprise-level DevOps automation, and high-availability system design were beyond the scope of this internship.

1.5. Report Organization

This report is organized into several chapters to clearly present the internship experience and technical contributions:

- **Chapter 1: Introduction** – Overview of the internship, objectives, scope, and problem statement.
- **Chapter 2: Organization Overview** – Introduction to Information Care Pvt. Ltd. and its working environment.
- **Chapter 3: Literature Review / Background Study** – Theoretical concepts related to full stack development and PERN stack technologies.
- **Chapter 4: Methodology and Tools Used** – Technologies and development workflow followed during the internship.
- **Chapter 5: System Design and Implementation** – Architecture, database design, API structure, and frontend implementation details.
- **Chapter 6: Results and Discussion** – Outcomes, challenges faced, and lessons learned.
- **Chapter 7: Conclusion and Future Recommendations** – Summary of experience and possible improvements.
- **Appendices** – Additional technical documentation and code references.

CHAPTER 2
ORGANIZATION DETAILS AND LITERATURE REVIEW

2.1. Introduction to Organization

Information Care Pvt. Ltd. is a Nepal-based IT company that provides software development and digital solutions to businesses and organizations ("Information Care Pvt. Ltd. – Where Ideas Become Softwares", 2026). The company focuses on delivering customized web applications, management systems, and digital platforms tailored to client requirements. It emphasizes practical, scalable, and user-friendly software systems that support business operations and digital transformation.

The organization works with modern web technologies and structured development methodologies to build reliable applications. Its services include web-based system development, database-driven applications, business management software, and technical consultation. The company aims to provide efficient and cost-effective software solutions while maintaining quality, security, and maintainability standards.

During my internship at Information Care Pvt. Ltd., I worked as a Software Development Intern in the full stack development team, where I was primarily involved in building and maintaining web applications using the PERN stack (PostgreSQL, Express.js, React.js, and Node.js).

Table 2.1: Internship Details

Organization Name	Information Care Pvt. Ltd.
Location	Kathmandu, Nepal
Internship Position	Software Development Intern
Internship Period	02/11/2025 - 02/02/2026

2.2. Organizational Hierarchy

Information Care Pvt. Ltd. follows a structured organizational model to ensure smooth coordination between technical and administrative departments. The general hierarchy includes:

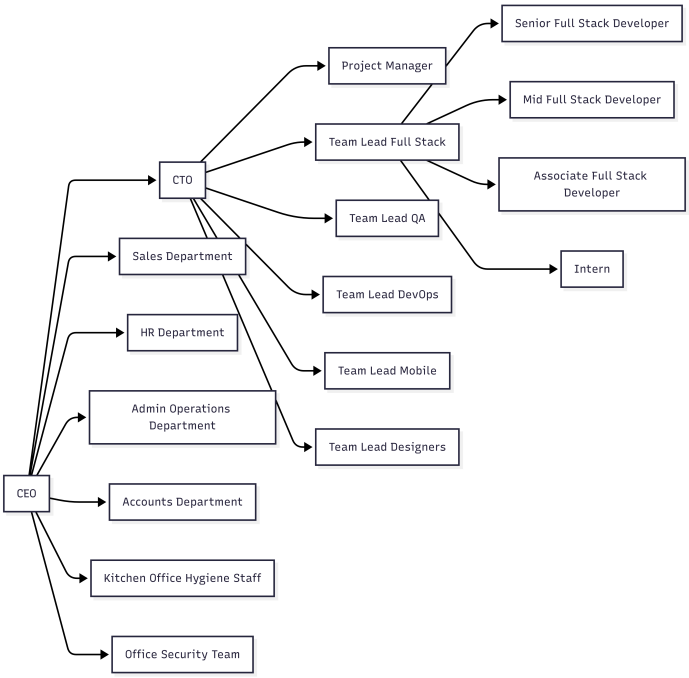

Figure 2.1: Organizational structure of Information Care Pvt. Ltd.

The development team works collaboratively under the guidance of senior developers and project managers. Tasks are assigned based on project requirements, and regular discussions are conducted to review progress, solve technical challenges, and maintain code quality standards.

2.3. Working Domains of Organization

Information Care Pvt. Ltd. operates in several key domains of software development and digital services. The primary working areas include:

- **Web Application Development:** Designing and developing dynamic, database-driven web applications for various business needs.
- **Custom Software Solutions:** Building tailored management systems such as inventory systems, administrative systems, and business automation tools.
- **Frontend and UI Development:** Creating responsive and interactive user interfaces using modern JavaScript frameworks.
- **Backend System Development:** Developing secure server-side logic, RESTful APIs, and database integration.
- **Database Design and Management:** Designing relational database schemas and optimizing queries for performance.

- **System Maintenance and Support:** Updating, debugging, and improving existing systems to ensure reliability and performance.
- **Website Development and Deployment:** Building and deploying business websites and web portals.

The company focuses on practical implementation, clean coding standards, and maintainable system architecture to ensure long-term sustainability of software solutions.

2.4. Description of Intern Department

During my internship, I was assigned to the Software Development Department, specifically working with the full stack development team. This department is responsible for designing, developing, testing, and maintaining web-based systems for clients.

My primary responsibilities included:

- Developing frontend interfaces using React.js
- Creating RESTful APIs using Node.js and Express.js
- Designing and managing PostgreSQL database schemas
- Implementing CRUD operations
- Integrating authentication and authorization using JWT
- Debugging and testing application features
- Maintaining version control using Git

The department follows a practical development workflow where frontend and backend components are developed in coordination. Regular reviews and feedback sessions helped improve code structure, performance, and security.

2.5. Literature Review

Full stack web development refers to the end-to-end design and implementation of web applications, encompassing both client-side (frontend) and server-side (backend) components. Modern web applications require a layered architecture that separates concerns into distinct layers: presentation layer (frontend UI), business logic layer (backend API and server logic), and data layer (database management) (Newman, 2015).

The PERN stack (PostgreSQL, Express.js, React.js, Node.js) represents a modern, unified approach to full stack development, enabling developers to use JavaScript across most application layers while leveraging PostgreSQL for robust relational database management.

Frontend Development: React.js

React.js is a component-based JavaScript library for building user interfaces. React employs declarative programming where developers describe the desired UI state, and React efficiently updates the DOM using its Virtual DOM mechanism. Component-based architecture promotes code reusability, maintainability, and testability. State management through React's hooks (useState, useEffect) and lifecycle methods enables functional components to manage complex interactions and side effects efficiently.

Backend Development: Node.js and Express.js

Node.js provides an event-driven, non-blocking I/O runtime that efficiently handles concurrent connections, making it ideal for I/O-intensive operations like database queries. Express.js is a lightweight web framework built on Node.js that simplifies API development through routing, middleware integration, and request/response handling.

RESTful API architecture, commonly implemented using Express.js, follows the representational state transfer (REST) principles that leverage standard HTTP methods and status codes for intuitive, scalable API design (Fielding, 2000). Express.js middleware implements the Chain of Responsibility pattern, where requests pass through composable functions for processing (logging, validation, authentication, compression).

Authentication and Security

Secure applications implement JSON Web Token (JWT) based authentication, which provides stateless token-based authentication suitable for distributed systems (Jones, 2015). JWTs are digitally signed tokens containing user claims, eliminating the need for server-side session storage and reducing session management overhead. Role-based access control (RBAC) further enhances security by restricting API endpoints based on user roles.

Input validation and error handling are critical for preventing injection attacks and malformed data. Server-side validation is essential as client-side validation can be bypassed. Secure password hashing with bcrypt or Argon2 and proper error handling prevent common vulnerabilities (McGraw, 2006).

Database Design: PostgreSQL

PostgreSQL is an advanced relational database management system known for reliability and ACID compliance (Atomicity, Consistency, Isolation, Durability). Database design using normalization principles reduces data redundancy and improves data integrity by organizing data into tables and

defining relationships through foreign keys (Codd, 1970).

Key design considerations include:

- **Normalization:** Decomposing tables to eliminate data duplication and maintain consistency.
- **Indexing and Query Optimization:** Using B-tree and specialized indexes to accelerate query execution.
- **ACID Compliance:** Ensuring transactions maintain data consistency and reliability even during failures.
- **Connection Pooling:** Managing multiple connections efficiently to improve application performance in concurrent scenarios.

Software Development Practices

Structured approaches to software development ensure quality and reliability. Agile methodologies emphasize iterative development, continuous feedback, and team collaboration through sprints and regular reviews (Beck, 2001). Git-based version control enables collaborative development, code reviews, and complete project history maintenance. Branching strategies like Git Flow allow parallel development while maintaining code quality.

Test-Driven Development (TDD) involves writing tests before implementation, ensuring code is testable and requirements are clear. Comprehensive testing through unit tests, integration tests, and end-to-end tests improves application reliability (Sommerville, 2010).

Continuous Integration and Continuous Delivery

CI/CD pipelines automate building, testing, and deployment processes. Continuous integration automatically tests code on each commit, detecting integration issues early. Continuous delivery automates deployment to staging environments, while continuous deployment automatically deploys to production (Humble & Farley, 2010). Automation reduces human error and accelerates feature delivery while maintaining code quality.

In summary, the PERN stack provides a cohesive platform for full stack web development. Component-based React architecture enables responsive, maintainable frontend development. Node.js and Express.js facilitate scalable backend development with well-designed REST APIs. PostgreSQL ensures reliable, normalized data management. Combined with secure authentication, proper validation, comprehensive testing, and CI/CD practices, developers can build reliable, scalable, and maintainable web applications that meet professional standards.

CHAPTER 3
INTERNSHIP ACTIVITIES

3.1. Roles and Responsibilities

During my internship at **Information Care Pvt. Ltd.**, I worked as a **Software Development Intern** in the full stack development team using the **PERN stack (PostgreSQL, Express.js, React.js, Node.js)**. My roles and responsibilities were aligned with structured software development practices and real-world business requirements.

- Designed and developed dynamic web application features using React.js for frontend development.
- Developed RESTful APIs using Node.js and Express.js following proper route structuring and middleware usage.
- Designed and normalized PostgreSQL database schemas to maintain data integrity and reduce redundancy.
- Implemented CRUD operations across frontend, backend, and database layers.
- Applied separation of concerns by maintaining clear distinction between presentation layer, business logic layer, and data layer.
- Implemented JWT-based authentication and role-based authorization mechanisms.
- Managed asynchronous operations in backend services using async/await and proper error handling techniques.
- Performed input validation and structured error handling to enhance application reliability.
- Optimized SQL queries and improved API response performance.
- Used Git for version control and collaborative development.
- Participated in debugging, feature testing, and system improvement tasks.

3.2. Weekly Log

The following table provides a weekly breakdown of activities and tasks completed during the 12-week internship:

Table 3.1: Weekly log of activities performed during the internship.

Week	Activity Performed
Week 1	Introduction to company workflow, project requirements analysis, and development environment setup (Node.js, PostgreSQL, React, Next.js).
Week 2	Understanding existing company website architecture and codebase; familiarization with React to Next.js migration strategy.
Week 3	Migration of company website from React to Next.js; implementation of server-side rendering and routing structure.
Week 4	Development of responsive frontend components for company website; implementation of modern UI design patterns.
Week 5	Creation of user-reactive components and interactive UI elements; optimization of component rendering performance.
Week 6	Introduction to Chiurika CMS project; requirement analysis and understanding of CMS architecture and functionality.
Week 7	Implementation of CMS features for Chiurika website; development of content management interfaces and dynamic content rendering.
Week 8	Database schema design for CMS content storage; integration of backend APIs with frontend CMS components.
Week 9	Implementation of authentication and authorization for CMS admin panel; role-based access control for content management.
Week 10	Integration and testing of complete CMS workflow; debugging and resolving frontend-backend integration issues.
Week 11	UI/UX enhancement and responsiveness testing across devices; performance optimization and code refactoring.
Week 12	Final testing, bug fixes, documentation preparation, and deployment support for both projects.

3.3. Description of the Project Involved During Internship

During my internship at **Information Care Pvt. Ltd.**, I was involved in two major projects: redesigning and migrating the company's official website from React to Next.js, and developing a Content Management System (CMS) for the Chiurika website.

Project 1: Company Website Redesign and React to Next.js Migration

The company website required modernization and migration from React.js to Next.js to leverage server-side rendering, improved performance, and SEO optimization. I analyzed the existing React codebase, planned the migration strategy, and executed the conversion of components to Next.js architecture. The migration involved converting client-side rendered pages to server-side rendered and static generated pages, implementing Next.js file-based routing, and optimizing the build process.

Following the migration, I focused on frontend development to create responsive and modern UI components. I designed and implemented reusable components with modern design patterns, ensuring consistent styling and functionality across all pages. Interactive UI elements were created using React hooks and state management, enabling dynamic user interactions. The components were optimized for performance, ensuring fast rendering and minimal re-renders. Cross-device testing was performed to ensure responsive design across desktop, tablet, and mobile platforms.

Project 2: Chiurika CMS Development

The Chiurika CMS project involved transforming the existing Chiurika website into a full-featured content management system. The project required designing and implementing a complete CMS workflow including content creation, editing, publishing, and management interfaces.

On the frontend, I developed the CMS admin panel with an intuitive user interface for content management. Key components included a content editor interface with rich text support, content listing and management pages, media upload and management functionality, and content preview and publishing features. The CMS was designed with role-based access control to restrict access to authorized users.

On the backend, I designed normalized PostgreSQL database schemas to store content, media files, and user information efficiently. RESTful APIs were implemented using Node.js and Express.js to handle CRUD operations for content management. Middleware functions were used for request validation, authentication verification, and structured routing. JWT-based authentication was implemented to secure the admin panel, ensuring that only authenticated users with appropriate roles could access content management features. Role-based authorization was applied to restrict specific operations based on user roles.

The development process followed structured software development lifecycle practices, including requirement analysis, system design, implementation, comprehensive testing, and deployment preparation. Version control using Git enabled collaborative development and maintained a complete history of changes.

3.4. Activities Performed

- **Website Migration and Architecture:** Analyzed existing React codebase, planned React to Next.js migration strategy, and executed component conversion with server-side rendering

implementation.

- **Responsive Frontend Development:** Designed and developed responsive React and Next.js components with modern UI patterns, ensuring consistent styling and functionality across all devices.
- **Interactive UI Components:** Created user-reactive components with dynamic behavior, form validation, user feedback mechanisms, and optimized component rendering performance.
- **CMS Admin Panel Development:** Implemented complete CMS admin interface including content editor, content listing, media management, and content publishing features.
- **Backend API Development:** Designed and implemented RESTful APIs using Node.js and Express.js with proper middleware for validation, authentication, and structured routing.
- **Database Design and Integration:** Designed normalized PostgreSQL schemas for CMS content storage, implemented database models and relationships, and optimized queries for performance.
- **Authentication and Authorization:** Implemented JWT-based authentication for secure user session management and role-based access control for restricting CMS operations.
- **Frontend-Backend Integration:** Integrated frontend CMS components with backend APIs, tested data flow, and resolved integration issues.
- **Testing and Debugging:** Performed functional and integration testing of complete workflows, identified and fixed bugs, and ensured system reliability.
- **Performance Optimization:** Optimized component rendering, API response times, SQL queries, and implemented code refactoring for maintainability.
- **Cross-Device Testing and Responsiveness:** Conducted responsive design testing across desktop, tablet, and mobile platforms to ensure consistent user experience.
- **Version Control and Collaboration:** Used Git for maintaining code history, managed code repositories, and facilitated collaborative development workflow.

CHAPTER 4
CONCLUSION AND LEARNING OUTCOMES

4.1. Conclusion

The internship at Information Care Pvt. Ltd. provided comprehensive practical exposure to full stack web development using the PERN stack (PostgreSQL, Express.js, React.js, Node.js) and modern web technologies including Next.js. Working on two significant projects—company website redesign and Chiurika CMS development—allowed me to understand how production-grade web applications are designed, developed, tested, and deployed in professional environments.

Throughout the 12-week internship period, I successfully completed the migration of the company website from React to Next.js, implementing server-side rendering for improved performance and SEO. I developed responsive and modern UI components, created interactive user-reactive elements, and optimized application performance across different devices. Additionally, I designed and implemented a complete Content Management System for the Chiurika website, including backend APIs, database architecture, user authentication, role-based authorization, and comprehensive admin interfaces.

The implementation of these projects significantly enhanced my understanding of modern web development practices, full stack architecture, component-based design patterns, and database normalization. I gained hands-on experience with frontend technologies (React.js, Next.js), backend frameworks (Node.js, Express.js), relational databases (PostgreSQL), and professional development tools (Git, version control).

This internship successfully bridged the gap between academic theory and industry practice, strengthening my competence in full stack development, system design, and professional software engineering. The experience enhanced my ability to solve real-world problems, design scalable and maintainable solutions, and contribute effectively within collaborative development environments. Overall, the internship was instrumental in developing my technical expertise as a Full Stack Web Developer and preparing me for professional roles in web development.

4.2. Learning Outcomes

During the internship, I achieved the following key learning outcomes:

- Developed practical understanding of React to Next.js migration strategies, including server-side rendering, static site generation, and file-based routing architecture.

- Gained hands-on experience with modern frontend technologies including React.js, Next.js, responsive design, and state management.
- Learned to design responsive and mobile-first UI components using modern CSS frameworks and design patterns.
- Strengthened knowledge of component-based architecture, reusable component design, and component composition patterns.
- Acquired experience in creating interactive user interfaces with dynamic behavior, form validation, and user feedback mechanisms.
- Developed practical skills in backend development using Node.js and Express.js for building scalable RESTful APIs.
- Learned to implement proper middleware patterns for request validation, error handling, and centralized authentication.
- Gained comprehensive understanding of database design using PostgreSQL, including schema normalization, relational constraints, and query optimization.
- Acquired hands-on experience implementing JWT-based authentication and role-based access control (RBAC) for secure applications.
- Learned full-stack integration practices, connecting frontend components with backend APIs and managing data flow across layers.
- Developed skills in database connection pooling, transaction management, and database performance optimization techniques.
- Gained practical experience with performance optimization including code splitting, component lazy loading, and query optimization.
- Strengthened knowledge of testing practices including functional testing, integration testing, and cross-browser/cross-device testing.
- Learned professional software development practices including Git version control, code review workflows, and collaborative development.
- Acquired experience in CMS architecture design, content management workflows, and admin panel development.
- Developed confidence in deploying and testing applications in staging and production environments.
- Enhanced professional skills such as team collaboration, technical documentation, time management, and communication.

- Gained practical understanding of the Software Development Lifecycle (SDLC) including planning, design, implementation, testing, and deployment.

  Overall, the internship significantly enhanced my technical expertise in full stack web development and professional competence, preparing me for careers as a Full Stack Web Developer or Software Engineer in modern web development environments.

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APPENDIX

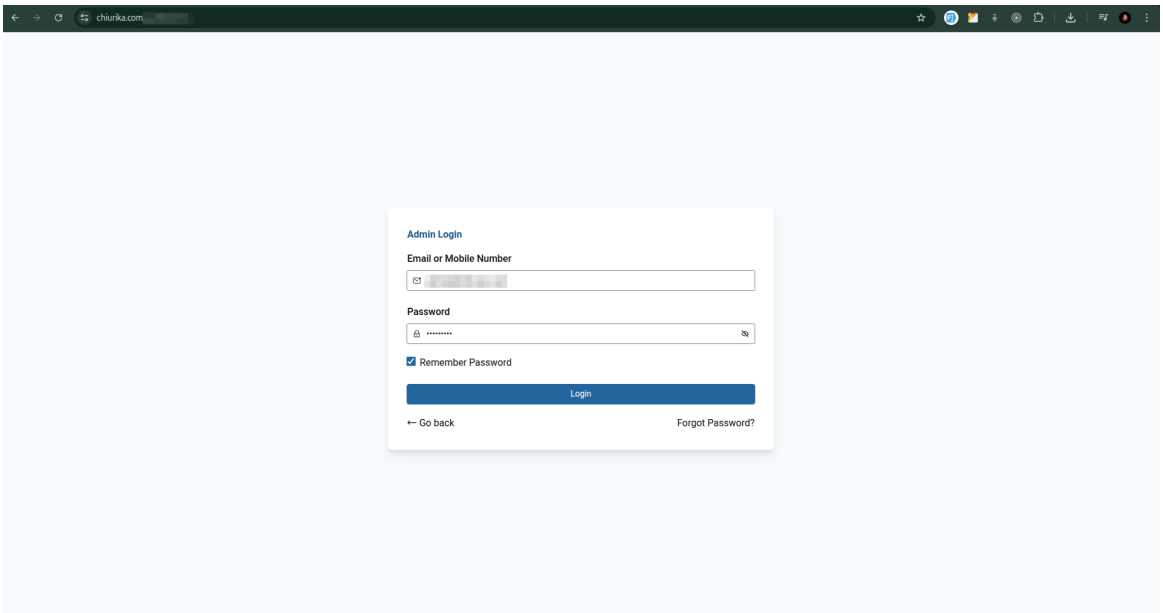

Figure 4.1: Chiurika Admin Login Page

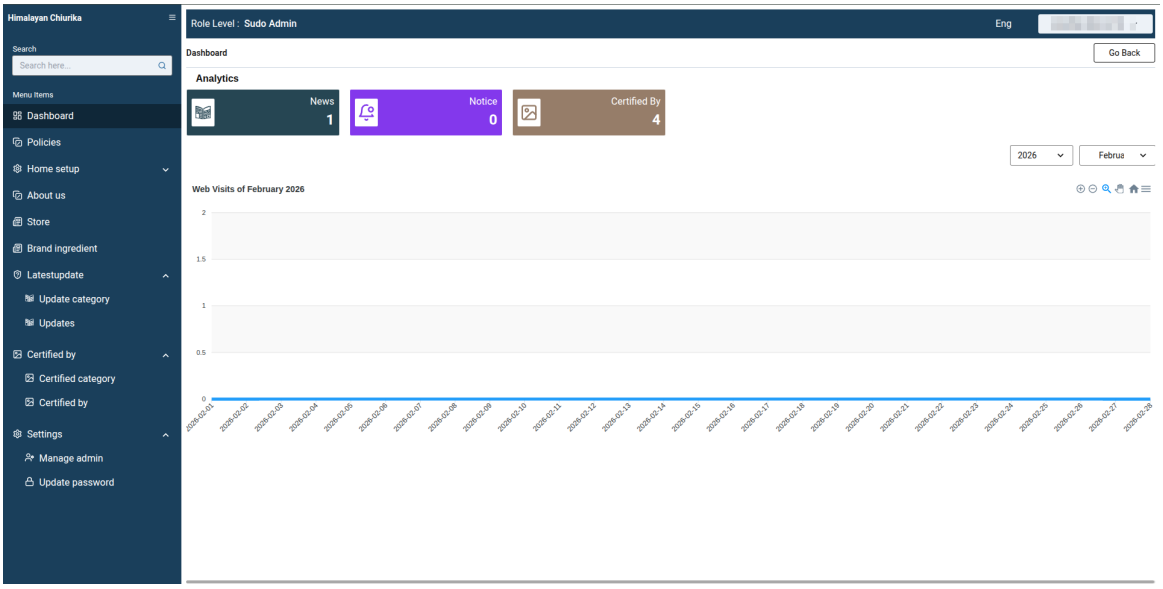

Figure 4.2: Chiurika Admin Dashboard

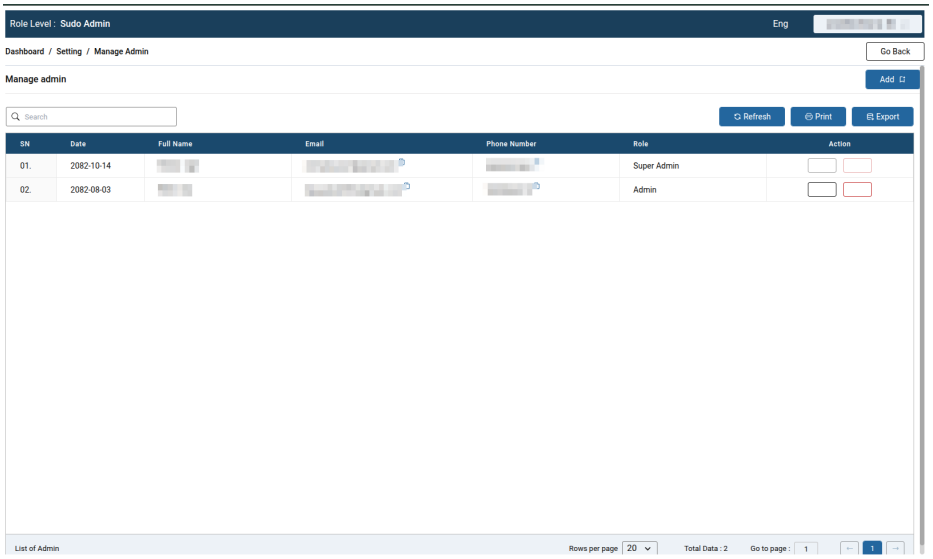

Figure 4.3: Manage Admin Page

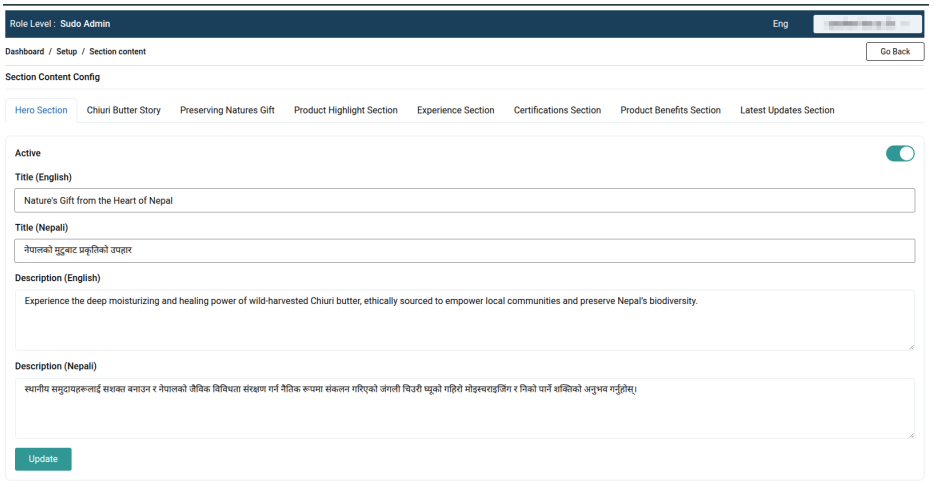

Figure 4.4: Content Management from Admin Panel

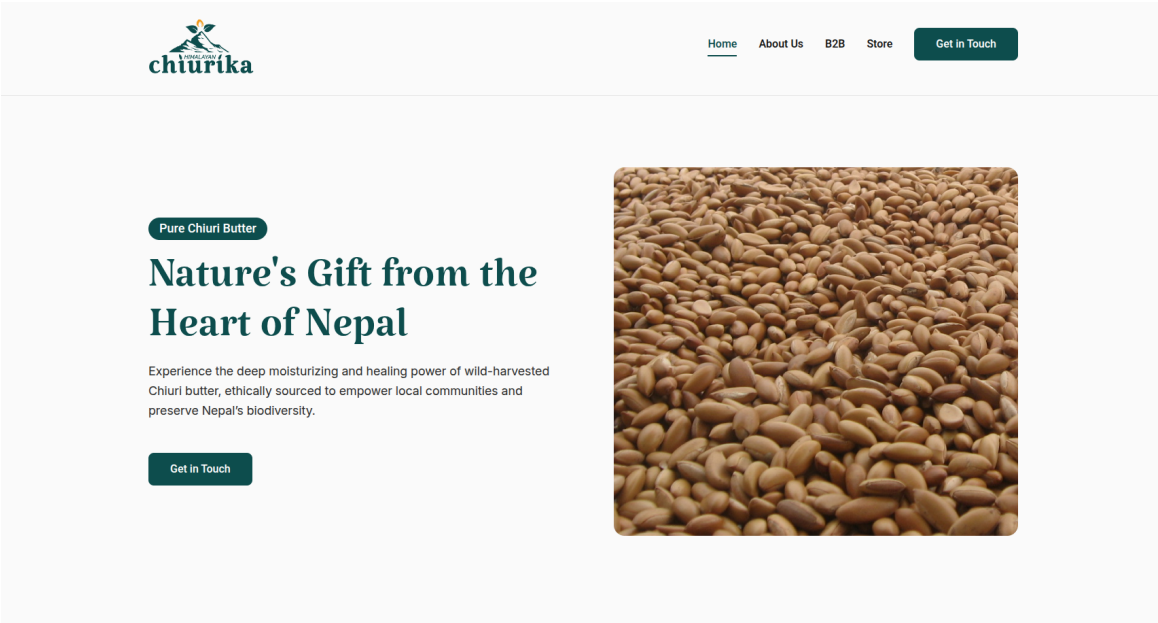

Figure 4.5: Updated content from admin panel to Website

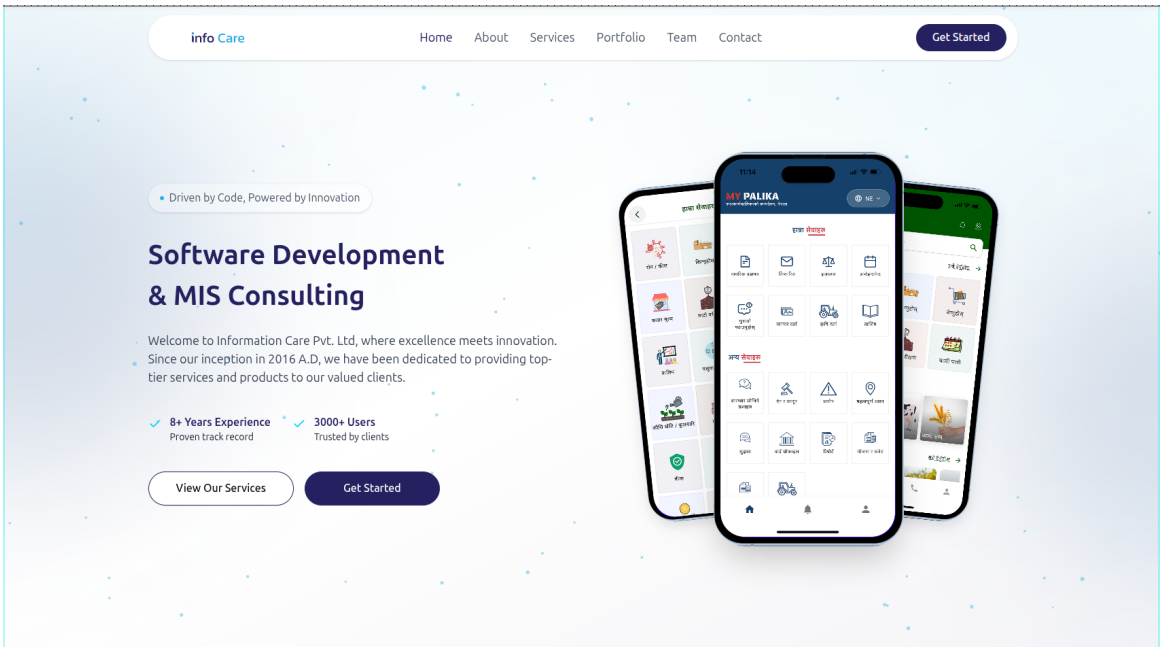

Figure 4.6: Official Information Care Website

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 1
Name of Student: Darshan Dhakal
Internship Site/ Organization: Information Care Pvt. Ltd
Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-07-16	9	9	Company orientation and introduction to existing projects.
2	2082-07-17	9	18	Setup development environment (Node.js, PostgreSQL, React, Next.js).
3	2082-07-18	9	27	Analyzed project requirements and company workflow documentation.
4	2082-07-19	9	36	Reviewed existing company website codebase and architecture.
5	2082-07-20	9	45	Studied React to Next.js migration best practices.
6	2082-07-21	Holiday	45	Weekend
7	2082-07-22	Holiday	45	Weekend

Current Week Summary: Completed onboarding and gained understanding of company projects and development environment setup.

Next Week Plan: Begin React to Next.js migration for company website.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 2

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-07-23	9	54	Analyzed existing React codebase structure and dependencies.
2	2082-07-24	9	63	Studied Next.js routing and server-side rendering concepts.
3	2082-07-25	9	72	Planned migration strategy and component conversion approach.
4	2082-07-26	9	81	Created Next.js project structure and configuration.
5	2082-07-27	9	90	Documented migration plan and component mapping.
6	2082-07-28	Holiday	90	Weekend
7	2082-07-29	Holiday	90	Weekend

Current Week Summary: Completed analysis of existing website and designed migration strategy from React to Next.js.

Next Week Plan: Start implementing website migration and component conversion.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 3

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-07-30	9	99	Migrated homepage and layout components to Next.js.
2	2082-07-31	9	108	Implemented server-side rendering for dynamic pages.
3	2082-08-01	9	117	Converted navigation and footer components.
4	2082-08-02	9	126	Setup Next.js routing structure and page organization.
5	2082-08-03	9	135	Tested migrated components and fixed routing issues.
6	2082-08-04	Holiday	135	Weekend
7	2082-08-05	Holiday	135	Weekend

Current Week Summary: Successfully migrated core components to Next.js with server-side rendering implementation.

Next Week Plan: Develop responsive and modern UI components.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 4

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-06	9	144	Implemented responsive design for home-page components.
2	2082-08-07	9	153	Created modern UI components with improved styling.
3	2082-08-08	9	162	Developed mobile-responsive navigation menu.
4	2082-08-09	9	171	Enhanced page layouts with modern design patterns.
5	2082-08-10	9	180	Optimized component rendering and CSS performance.
6	2082-08-11	Holiday	180	Weekend
7	2082-08-12	Holiday	180	Weekend

Current Week Summary: Developed responsive and modern UI components with improved design aesthetics.

Next Week Plan: Create user-reactive components and interactive features.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 5

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-13	9	189	Implemented interactive UI elements and animations.
2	2082-08-14	9	198	Created user-reactive components with dynamic behavior.
3	2082-08-15	9	207	Added form validation and user feedback mechanisms.
4	2082-08-16	9	216	Optimized component re-rendering and state management.
5	2082-08-17	9	225	Finalized interactive features and performance improvements.
6	2082-08-18	Holiday	225	Weekend
7	2082-08-19	Holiday	225	Weekend

Current Week Summary: Implemented user-reactive components with interactive features and optimized performance.

Next Week Plan: Begin Chiurika CMS project analysis and planning.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 6

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-20	9	234	Introduction to Chiurika CMS project requirements.
2	2082-08-21	9	243	Analyzed existing Chiurika website structure and content.
3	2082-08-22	9	252	Researched CMS architecture and functionality requirements.
4	2082-08-23	9	261	Designed content management interface wireframes.
5	2082-08-24	9	270	Created project documentation and implementation plan.
6	2082-08-25	Holiday	270	Weekend
7	2082-08-26	Holiday	270	Weekend

Current Week Summary: Completed requirement analysis and architecture design for Chiurika CMS project.

Next Week Plan: Implement CMS features and content management interfaces.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 7

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-08-27	9	279	Implemented CMS admin panel layout and navigation.
2	2082-08-28	9	288	Created content editor interface with rich text support.
3	2082-08-29	9	297	Developed content listing and management components.
4	2082-08-30	9	306	Implemented media upload and management functionality.
5	2082-08-31	9	315	Added content preview and publishing features.
6	2082-09-01	Holiday	315	Weekend
7	2082-09-02	Holiday	315	Weekend

Current Week Summary: Successfully implemented core CMS features including content editor and media management.

Next Week Plan: Design database schema and integrate backend APIs.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

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Weekly Internship Logbook

Week No: 8

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-03	9	324	Designed PostgreSQL database schema for CMS content storage.
2	2082-09-04	9	333	Implemented database models and relationships.
3	2082-09-05	9	342	Created RESTful APIs for content CRUD operations.
4	2082-09-06	9	351	Integrated frontend CMS components with backend APIs.
5	2082-09-07	9	360	Tested API endpoints and data flow validation.
6	2082-09-08	Holiday	360	Weekend
7	2082-09-09	Holiday	360	Weekend

Current Week Summary: Designed database schema and successfully integrated backend APIs with frontend CMS components.

Next Week Plan: Implement authentication and authorization for CMS.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

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Weekly Internship Logbook

Week No: 9

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-10	9	369	Implemented JWT-based authentication for admin panel.
2	2082-09-11	9	378	Created user login and session management system.
3	2082-09-12	9	387	Implemented role-based access control for content management.
4	2082-09-13	9	396	Added user authorization middleware and protected routes.
5	2082-09-14	9	405	Tested authentication flow and security mechanisms.
6	2082-09-15	Holiday	405	Weekend
7	2082-09-16	Holiday	405	Weekend

Current Week Summary: Successfully implemented authentication and role-based authorization for CMS admin panel.

Next Week Plan: Perform integration testing and UI enhancement.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 10

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-17	9	414	Conducted integration testing of complete CMS workflow.
2	2082-09-18	9	423	Debugged frontend-backend integration issues.
3	2082-09-19	9	432	Fixed content rendering and API response issues.
4	2082-09-20	9	441	Resolved authentication and authorization bugs.
5	2082-09-21	9	450	Documented resolved issues and system improvements.
6	2082-09-22	Holiday	450	Weekend
7	2082-09-23	Holiday	450	Weekend

Current Week Summary: Completed integration testing and resolved critical bugs in CMS functionality.

Next Week Plan: Enhance UI/UX and perform cross-device testing.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 11

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-09-24	9	459	Enhanced CMS admin panel UI with improved design.
2	2082-09-25	9	468	Improved responsiveness of CMS components across devices.
3	2082-09-26	9	477	Conducted cross-device testing (desktop, tablet, mobile).
4	2082-09-27	9	486	Optimized component performance and load times.
5	2082-09-28	9	495	Refactored code for better maintainability.
6	2082-09-29	Holiday	495	Weekend
7	2082-09-30	Holiday	495	Weekend

Current Week Summary: Enhanced UI/UX of both projects with improved responsiveness and performance optimization.

Next Week Plan: Final testing, deployment support, and documentation.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13

Central Campus of Technology

Dharan - 14

Weekly Internship Logbook

Week No: 12

Name of Student: Darshan Dhakal

Internship Site/ Organization: Information Care Pvt. Ltd

Address: Mid-Baneshwor, Kathmandu, Nepal

S.N.	Date	Hours Worked	No. of Hours Worked to Date	Activity
1	2082-10-01	9	504	Performed final testing of company website and Chiurika CMS.
2	2082-10-02	9	513	Fixed minor bugs and UI inconsistencies.
3	2082-10-03	9	522	Assisted in deployment preparation and environment setup.
4	2082-10-04	9	531	Conducted pre-deployment testing and verification.
5	2082-10-05	9	540	Completed documentation and internship wrap-up.
6	2082-10-06	Holiday	540	Weekend
7	2082-10-07	Holiday	540	Weekend

Current Week Summary: Completed final testing, bug fixes, and deployment support. Finalized all documentation and project handover.

Next Week Plan: Internship completed.

Total Hours Worked: 45

Student's Signature: _____ **Date:** 2082-10-07

Mentor's Signature: _____ **Date:** 2082-10-07

Supervisor's Signature: _____ **Date:** 2082-11-13